
TROPIC01

ODR_TR01_PSR_005_Decap_ZCU

Decapsulation of TROPIC01

Version: 1.0

Abstract

This security report provides the results of the TROPIC01 evaluation performed by the Department of Materials and Technology, UWB in Pilsen (KET ZČU). KET ZČU performed a study of the TROPIC01 packaging and provided a comprehensive report on the feasibility of the TROPIC01 decapsulation process from both sides.

June 3, 2026





Summary of the Security Report

Report Details

- **Evaluation Laboratory:** Department of Materials and Technology, UWB in Pilsen (KET ZČU)
- **Evaluation Laboratory Contact:** Ing. Robert Vik, Ph.D., rvik@fel.zcu.cz
- **Contract Type:** Paid Contract
- **Evaluation Period:** Multiple incremental evaluations within a 1-year period (07/2025 – 05/2026)
- **Type of Evaluation:** All TROPIC01 versions, TROPIC01 standard packages
- **Tropic Square Contact:** Tropic Square Security Team, security@tropicsquare.com

Tropic Square Executive Summary

Department of Materials and Technology, UWB in Pilsen (KET ZČU) conducted a study of the TROPIC01 packaging and provided a comprehensive report on the feasibility of the TROPIC01 decapsulation process from both sides.

Chip decapsulation is needed for security evaluations when access to the chip surface is required. Detailed knowledge of the complexity and repeatability of the decapsulation process with accessible equipment also helps assess the severity of attacks for which decapsulation of the chip is a prerequisite.

The report provides an overview of the TROPIC01 packaging and thoroughly describes the difficulties and challenges related to the TROPIC01 decapsulation process. The report also sets a reasonable boundary for the cost of ownership of the equipment allowing reliable (high confidence of success) chip decapsulation. The report might also serve security researchers when estimating the cost of decapsulation for chips similar to the TROPIC01 by analogy.

Chips decapsulated under the framework of this report have already been used for security evaluations by another Tropic Square partner lab.



Tropic Square Recommendations

The report sets a reasonable boundary for the cost of ownership of equipment allowing reliable (high confidence of success) chip decapsulation, which might be a prerequisite for certain high-complexity attacks on the TROPIC01. This report might be helpful in assessing the severity of such attacks in the context of applications using TROPIC01 or a similar chip.

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Version History

Version	Date	Description
1.0	3.6.2026	Initial Public Report Version



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The Partner Evaluation Report Follows

Decapsulation of TROPIC01 Chip

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Abstract

This report describes a methodology for the topside and backside decapsulation of the TROPIC01 integrated circuit to enable advanced security testing. To provide access to the chip while maintaining full electrical connectivity, a dedicated TROPIC01_LFI test board was designed. The process employs a hybrid approach, combining fiber laser ablation for bulk material removal with selective chemical etching to expose the silicon die, followed by manual mechanical removal of the die-attach adhesive the silicon while preserving surface integrity. Surface quality was verified using optical and confocal microscopy, confirming the successful removal of all materials without inducing mechanical or thermal damage. The resulting methodology provides a highly reliable and repeatable procedure for preparing decapsulated samples in failure-analysis and hardware-security research environments.

Keywords

TROPIC01, Decapsulation, Laser Ablation, Chemical Etching, Failure Analysis (FA), Microscopy, Hardware Security, PCB Design

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1 Introduction

This report documents the procedure and results of a specialized decapsulation of the TROPIC01 chip, performed by the Faculty of Electrical Engineering at the University of West Bohemia for Tropic Square s.r.o. The objective of the project was to execute both topside and backside decapsulation of the package while maintaining maximum die integrity and functionality for subsequent security analyses.

1.1 Project Objectives

The primary objective of this work is to develop a reliable and repeatable methodology for gaining access to the silicon die. The focus is on identifying an effective process that ensures die exposure without inducing thermal or mechanical degradation, while preserving the full electrical integrity of the package pins. This ensures the device remains fully functional for subsequent analysis from both the top and backside.

Three fundamental tasks were defined:

1. To establish a dependable procedure for accessing the active circuitry from the topside of the package and to provide a comprehensive description of the optimized workflow.
2. To develop a successful workflow for backside decapsulation by integrating laser and chemical techniques, ensuring the process is safe for the silicon die and consistently reproducible.
3. To design a test PCB for the TROPIC01 device, featuring an aperture beneath the component. This aperture must allow a test probe to be positioned as close as possible to the die's back surface while the component remains fully soldered to the electrical circuit.

2 Characterization of the TROPIC01 chip

The TROPIC01 device was supplied in a QFN32 package (4×4 mm) with a 0.4 mm lead pitch. The encapsulation consists of a standard silica-filled epoxy molding compound (EMC).

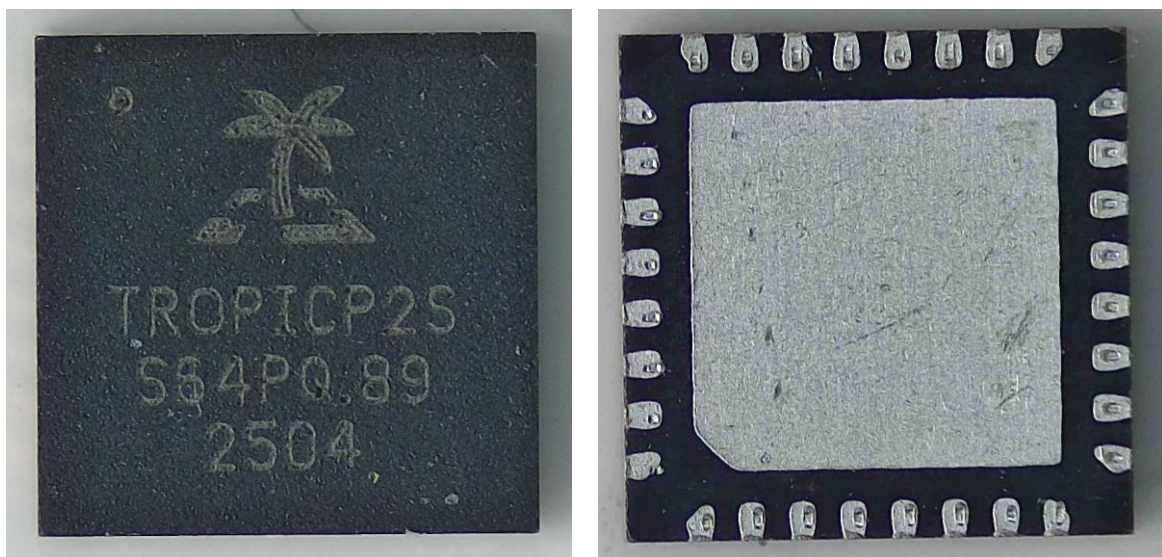


Figure 1 Top and bottom view of the TROPIC01 QFN32 package (4×4 mm).

To evaluate the material composition and technological topology of the device, X-ray imaging (Fig. 2) and metallographic cross-sectioning (Fig. 3) were performed. These steps were used to accurately map the geometry and thickness of the individual package layers. The silicon die, measuring approximately 1.85×2.7 mm, is centered within the package and is connected to the leads via copper wire bonds.

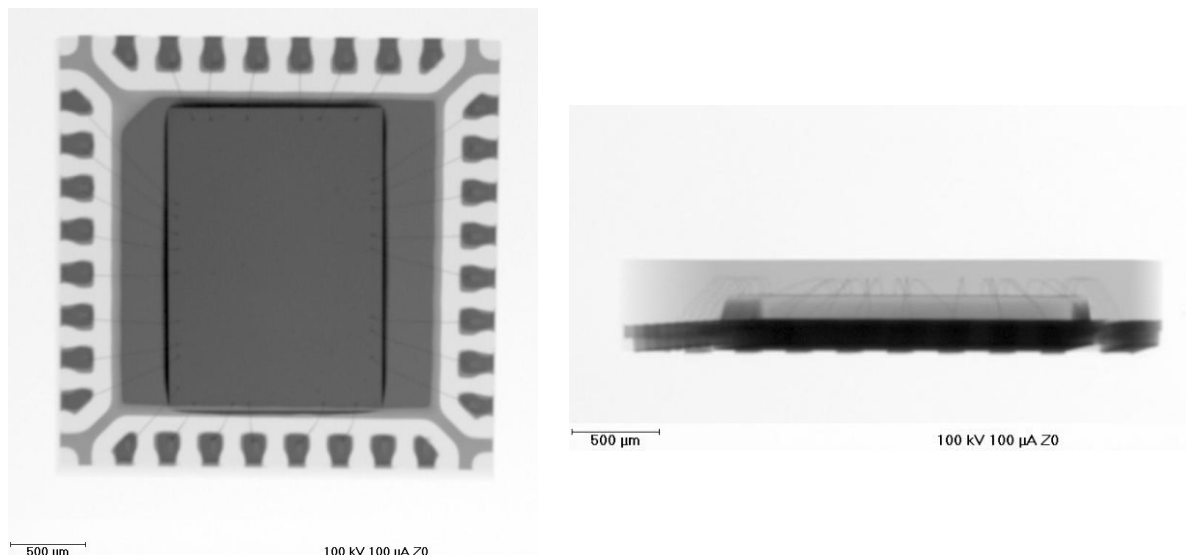


Figure 2 X-ray image of the TROPIC01 package showing the die placement within the encapsulation.

X-ray analysis revealed that the wire bonds are located approximately 100 μm from the top surface of the package. A copper thermal pad with a thickness of approximately 200 μm is situated on the bottom side of the package.



Figure 3 Metallographic cross-section of the TROPIC01 package.

3 Design of test PCB

The TROPIC01_LFI test board was designed for testing decapsulated TROPIC01 chip packages from both the top and backside. The board layout is based on the LFI interface, utilizing two 1×20-pin connectors with a 2.54 mm pitch. Additionally, a mikroBUS-compliant interface for SPI communication and a dedicated header for the Tropic USB Stick devKIT have been integrated. The board features solder bridges JP1 through JP12, which are connected by 0.2 mm PCB traces in the default configuration. For testing purposes requiring signal disconnection, these traces must be manually removed (cut).

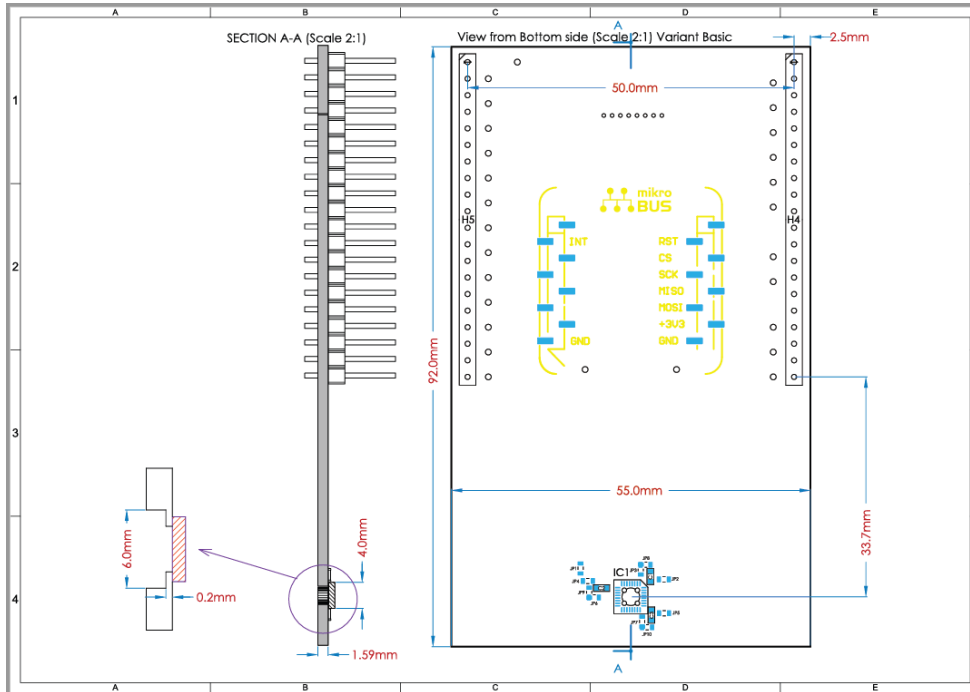


Figure 4 Dimensional drawing of the TROPIC01_LFI board.

The PCB is fabricated from high- T_g FR4 substrate material featuring a black solder mask. To provide sufficient clearance for testing, the chip is positioned 30 mm away from the connector axis. For backside analysis, a milled aperture is located directly beneath the chip package to allow direct access to the die. The milling process was controlled to remove material such that the residual PCB thickness beneath the mounted chip is only 200 μm .



Figure 5 The top and bottom sides of the TROPIC01_LFI board, showing the mounted chip and the backside access window.

4 Methodology and Technical Implementation

The decapsulation of the TROPIC01 integrated circuit was performed using a combined physical-chemical procedure. This approach utilizes laser ablation to remove the bulk of the molding compound, followed by selective chemical etching using an acid mixture to expose the die surface.

4.1 Topside Decapsulation

The procedural workflow selected for the topside decapsulation of the TROPIC01 device commenced with laser-assisted cavity formation, followed by the application of a concentrated acid mixture directly into the cavity. Once the reaction progressed, it was terminated and neutralized, allowing for a thorough inspection of the cavity, the die surface, and the integrity of the wire bonds. This sequence of acid application, termination, and inspection was repeated as required until the active die surface was fully exposed.

This general procedure is standard for wet-etch decapsulation of components with gold wire bonds, typically using fuming nitric acid (HNO_3 , concentration > 90 %). However, for copper wire bonds, the process must be modified since copper, unlike gold, dissolves rapidly in nitric acid. Therefore, concentrated sulfuric acid (H_2SO_4) is added to the HNO_3 to facilitate the formation of a protective passivation layer of bisulphate anions on the copper surface. The mixture is typically prepared in a 3:1 or 2:1 ratio

($\text{HNO}_3\text{:H}_2\text{SO}_4$) and operated at elevated temperatures (60–90 °C) to enhance etch activity and minimize the chemical exposure time of the bonds.

Specific topside decapsulation procedure is as follows:

1. A 3×2.8 mm cavity with a depth of approximately 100 µm (reaching the upper bond wire loops) was created using a fiber laser. The laser power was set to 40 % (8 W) at a scan speed of 10,000 mm/s. The ablation depth was monitored via microscopy (Fig. 4).
2. The PCB carrying the chip was preheated on a hot plate at 60 °C for 5 minutes. The acid mixture $\text{HNO}_3\text{:H}_2\text{SO}_4$, 3:1 ratio) was simultaneously heated to the same temperature.
3. Using a micropipette, 0.5 µl of the heated acid mixture was applied to the cavity and allowed to react for 15 seconds. Precise volume control is critical to prevent reaction off-gassing from causing an acid overflow, which would corrode the external package leads.
4. After 15 seconds, the reaction was quenched, and the resulting mold compound residue was rinsed with acetone. The board was then placed in a beaker containing acetone, and the beaker was immersed in a water-filled ultrasonic bath for 10 seconds of sonication.
5. The condition of the die and bonds was inspected under a microscope.
6. Steps 3–5 were repeated until the die surface was completely cleared.
7. The board was then placed in a beaker containing distilled water and sonicated in the ultrasonic water bath for 1 minute to remove residual acids and salts.
8. A final rinse was performed using isopropyl alcohol (IPA), and the surface was dried with compressed air.

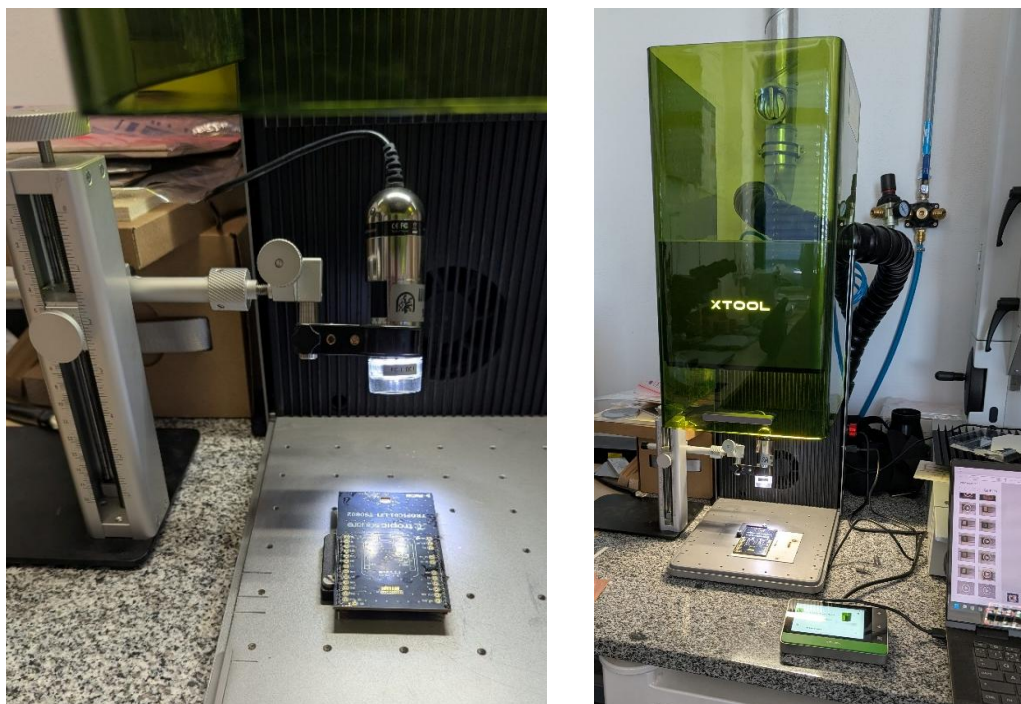


Figure 6 Experimental setup for laser ablation of the molding compound.

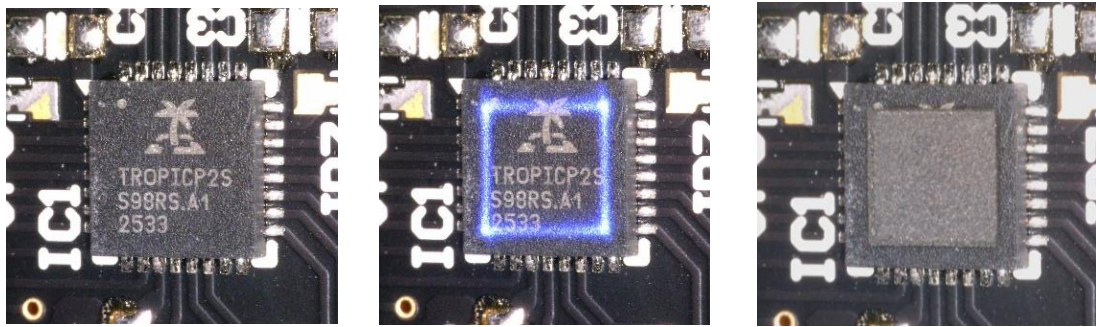


Figure 7 Stages of package laser ablation during topside decapsulation.

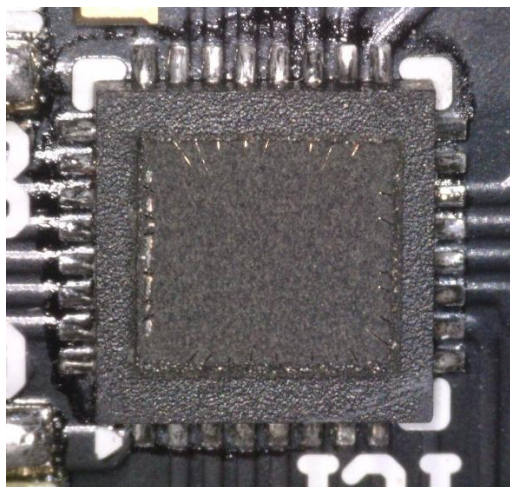


Figure 8 Detail of the 100 μm deep cavity with partially exposed wire bonds.



Figure 9 Sample preparation for the chemical decapsulation phase.

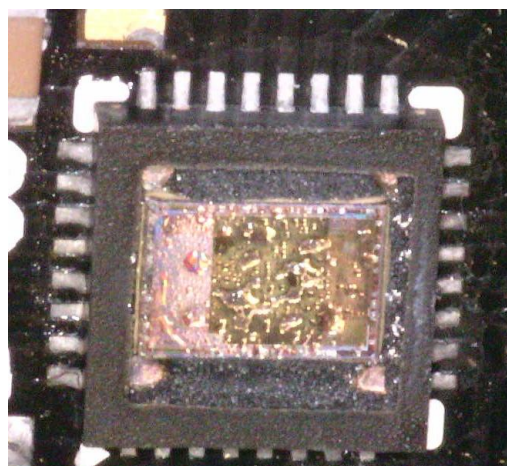
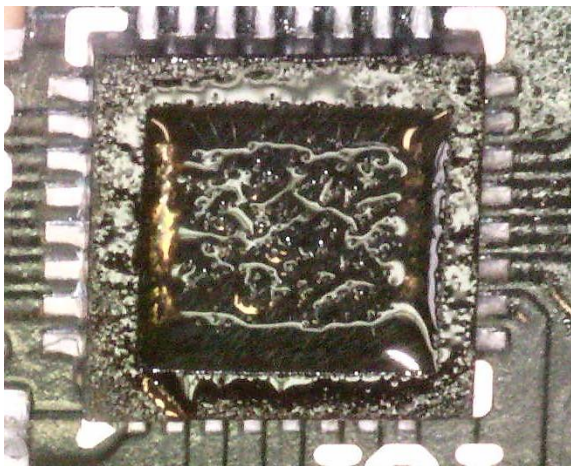


Figure 10 Selective etching process using a 3:1 $\text{HNO}_3:\text{H}_2\text{SO}_4$ mixture at 60 °C.

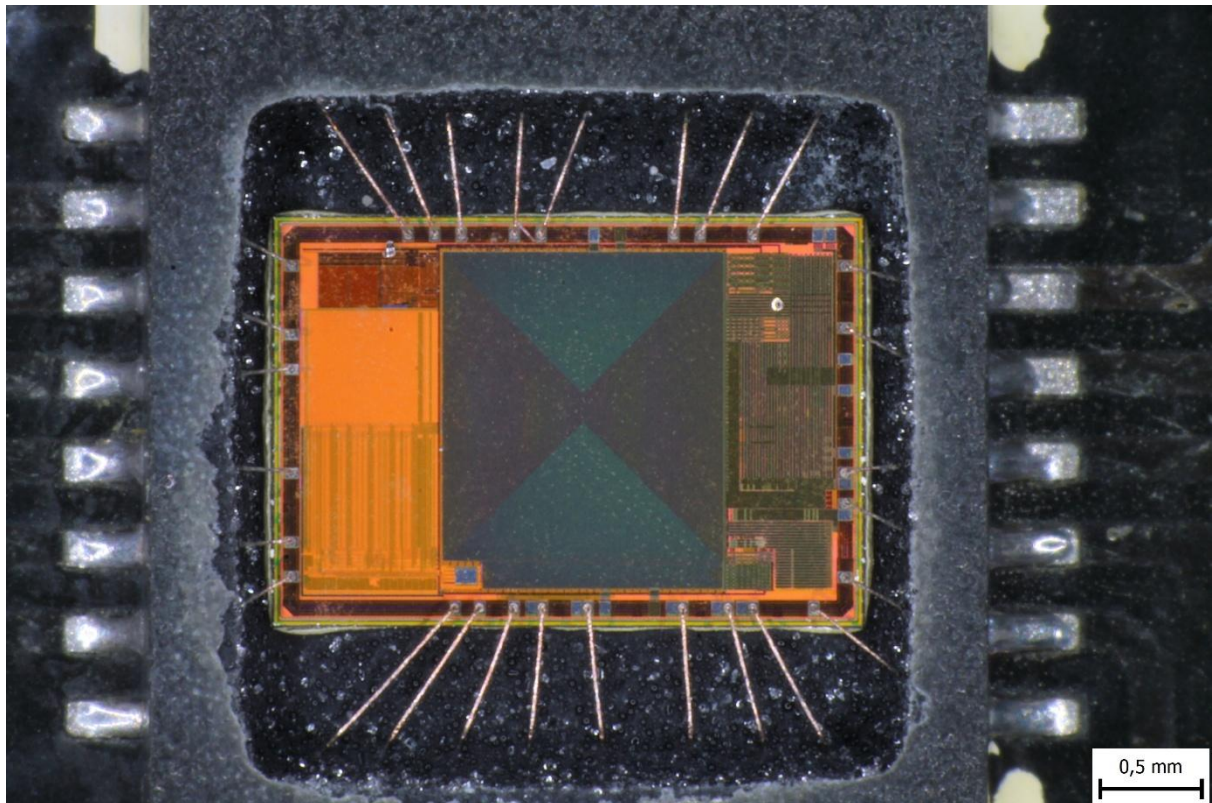


Figure 11 Final state of the device after decapsulation and post-process cleaning.

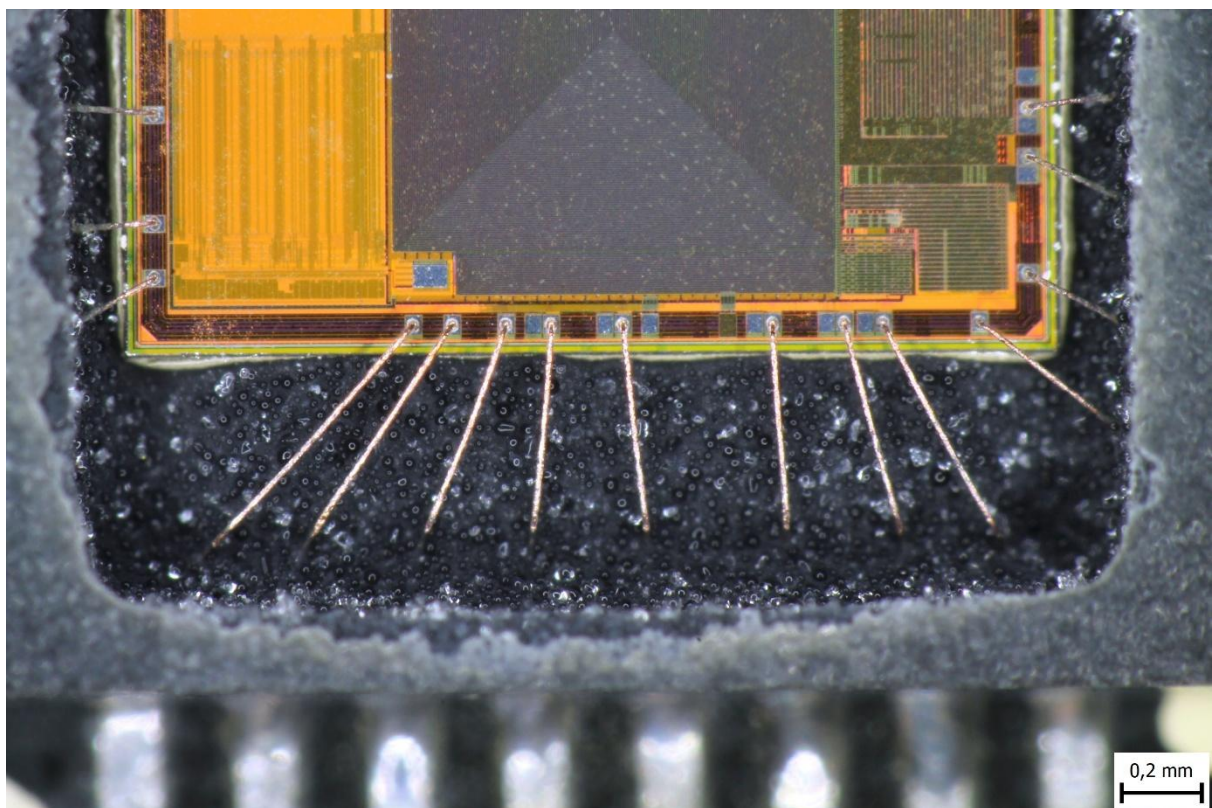


Figure 12 Close-up view of the wire bond condition following completion of decapsulation and final cleaning.

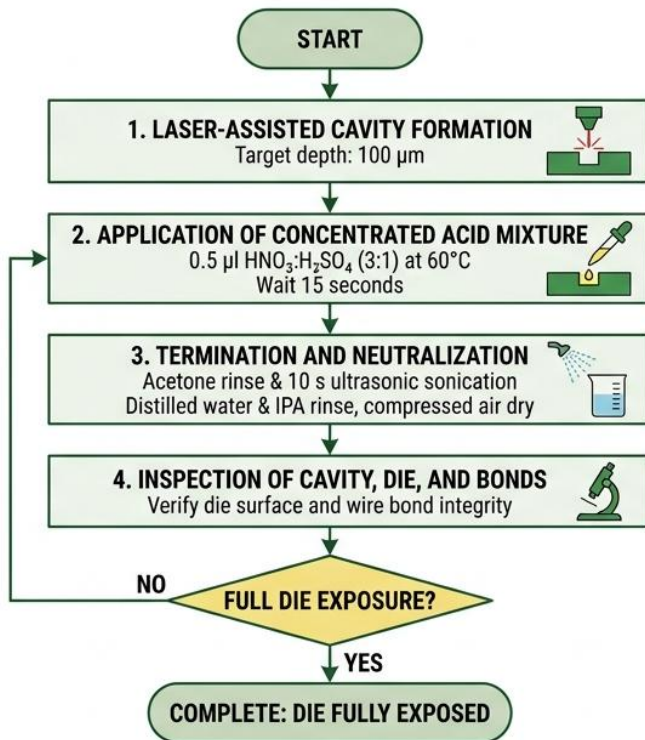


Figure 13 Process flow diagram for topside decapsulation.

4.2 Backside Decapsulation

The second task involved gaining access to the silicon die surface from the backside of the package to enable security testing via targeted IR laser irradiation on a functional, mounted device. This required the design of a specialized test PCB with an aperture aligned with the device's thermal pad, providing optical access to the die while the component remained soldered to the opposite side of the board.

Backside exposure is significantly more technically challenging than topside decapsulation. Analysis showed that the die is mounted to a copper thermal pad (approx. 200 µm thick) using a conductive die-attach epoxy. Laser ablation of the copper required full power and numerous passes, raising concerns regarding potential thermal damage due to heat accumulation. Furthermore, the ablation profile was non-uniform; the laser penetrated deeper at the cavity perimeter, leading to localized silicon destruction even while the center remained covered by copper.

Consequently, a hybrid approach was adopted using selective chemical etching with 65 % nitric acid (HNO₃). At this concentration, the acid reacts rapidly with copper but remains inert toward the molding compound and the die-attach epoxy, ensuring that only the metallic material within the cavity is dissolved. To prevent acid migration to the solder joints, a layer of soldering flux was applied and reflowed before decapsulation, creating an organic protective barrier at the PCB-to-package interface.

Attempts to remove the residual epoxy layer via laser ablation resulted in surface damage to the silicon. Therefore, the epoxy was manually removed using a custom-made copper scraper mounted in a mechanical lead holder (clutch pencil).

Specific Backside Decapsulation Procedure:

1. Following the soldering of the chip to the PCB, flux was applied to the top side and reflowed using hot air to create a protective chemical barrier.
2. The PCB was positioned in the laser system with the aperture facing upward. A fiber laser was used to mill an initial cavity into the copper thermal pad with dimensions of 2.8×2.7 mm and a target depth of 150 μm .
3. A 1 μl droplet of nitric acid (65 % HNO_3) was applied to the cavity at room temperature. The reaction produced a blue-green copper(II) nitrate solution; as the concentration of the reaction products increased, the etch rate decreased significantly.
4. The reaction byproduct was removed by immersing the board in a beaker of deionized water, followed by drying the cavity with a stream of compressed air.
5. Steps 3 and 4 were repeated until the copper thermal pad was fully dissolved, exposing the conductive die-attach epoxy layer (Fig. 17).
6. The board was secured in a vise, and the die-attach epoxy on the backside of the chip was manually scraped off under a microscope using a custom-made scraper (Fig. 19).
7. The board was placed in a beaker with isopropyl alcohol (IPA) and sonicated in an ultrasonic water bath for 60 seconds.
8. The exposed die backside was rinsed with acetone and dried using a stream of compressed air.

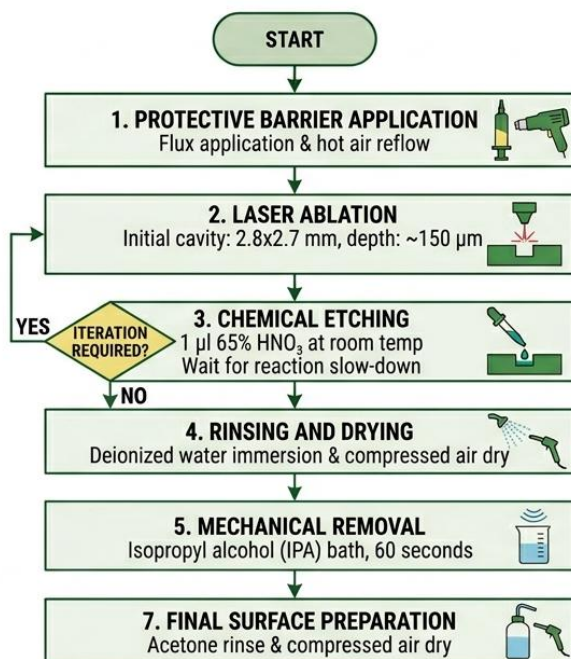


Figure 14 Process flow diagram for backside decapsulation.

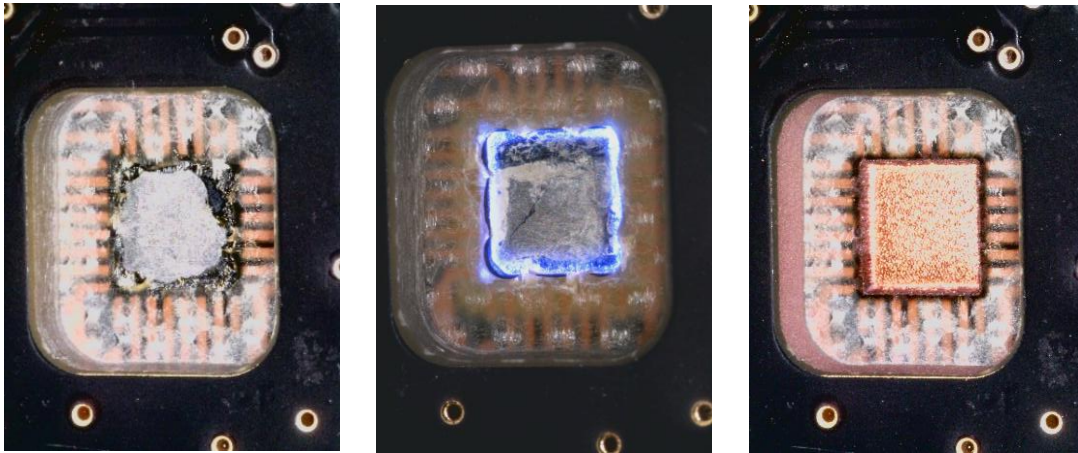


Figure 15 Stages of copper thermal pad laser ablation during backside decapsulation.

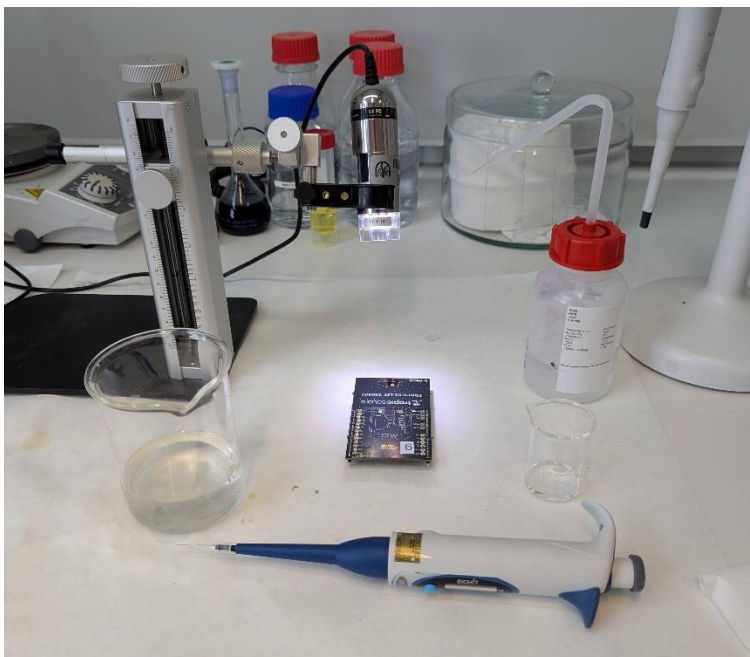


Figure 16 Sample preparation for the Cu pad chemical etching.

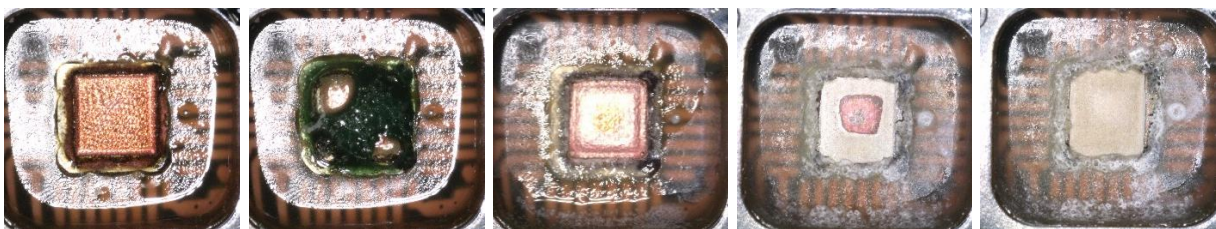


Figure 17 Iterative chemical etching of copper thermal pad using 65% HNO_3 .

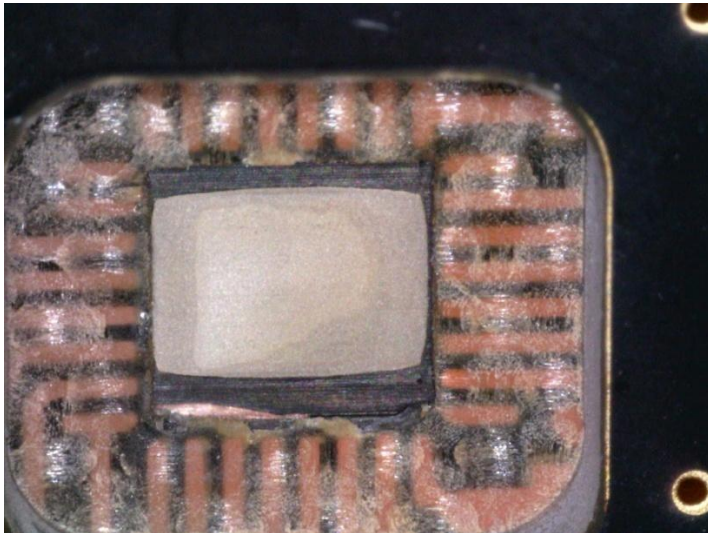


Figure 18 Cavity inspection following selective etching of the thermal pad, revealing the exposed conductive die-attach adhesive on the backside of the die.

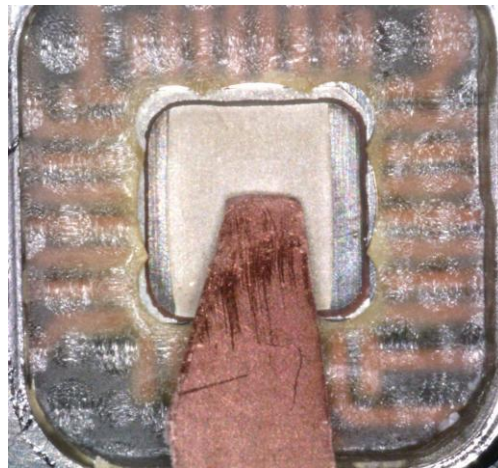
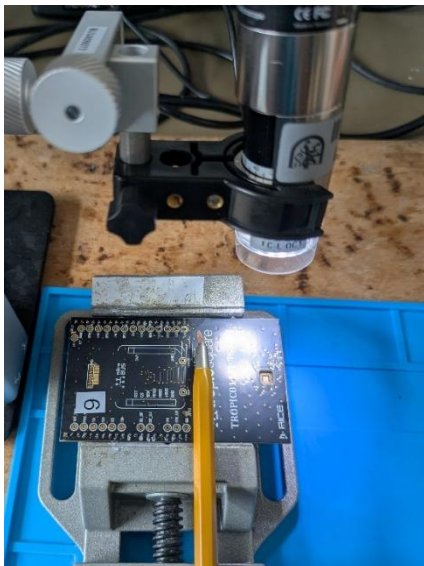


Figure 19 Manual scraping of die-attach adhesive using a custom-made scraper.

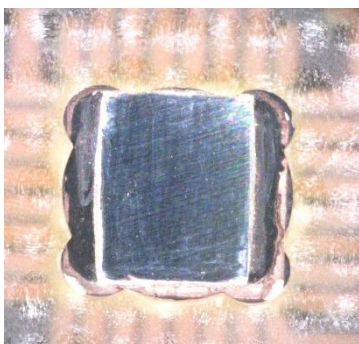


Figure 20 Detailed view of the exposed silicon die backside following the completion of the backside decapsulation process.

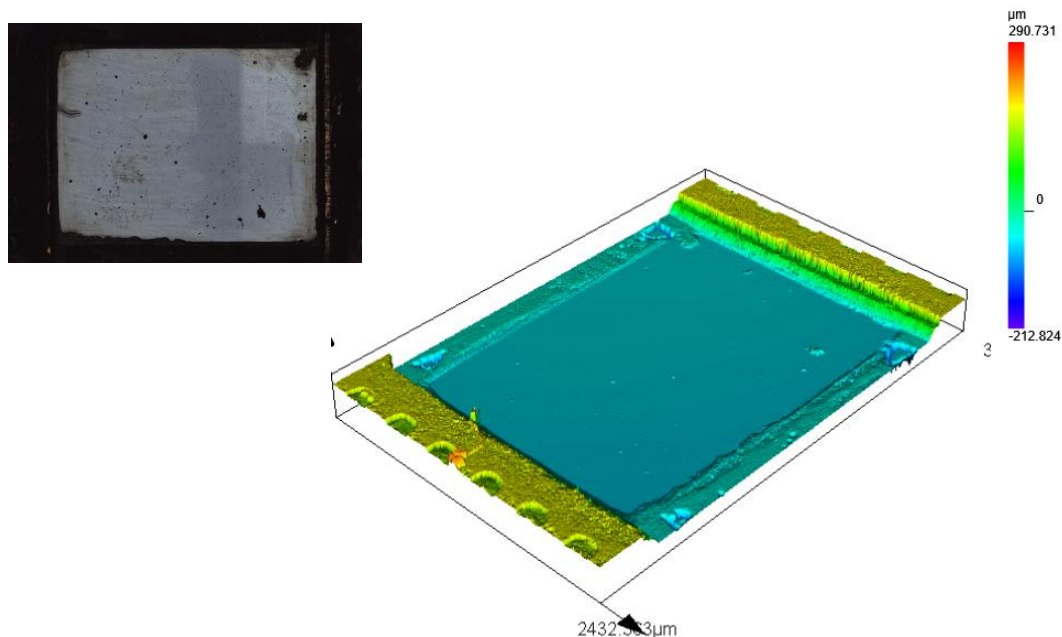


Figure 21 Surface integrity verification of the exposed die backside using confocal microscopy following the complete removal of the die-attach adhesive.

4.3 Technical-Economic Evaluation and Alternative Approaches

The equipment and reagents used in both decapsulation procedures were as follows:

- *Laser system*: xTool F1 Ultra (20W Fiber laser) – approx. €3,000
- *Laboratory hot plate*: IKA RCT basic (2022 model) – approx. €1,000
- *Micropipette*: Biohit mLine 0.5–10 µl – approx. €500
- *Digital microscope*: Dino-Lite AD7013MTL – approx. €1000
- *Ultrasonic cleaner*: RK 510 H – approx. €1800
- Fuming nitric acid (HNO_3 , > 90 %), 500 ml – approx. €250
- Sulfuric acid (H_2SO_4 , 95–98 %), 500 ml – approx. €80
- Cleaning solvents: Acetone, Isopropyl alcohol (IPA) – approx. €10
- Soldering flux CIF F42202 – €20

The total acquisition cost for the instrumentation and reagents required to implement the complete decapsulation process is approximately **€7,660**. The costs listed above represent the approximate purchase prices of the specific laboratory instruments used in this study. The described process should not be considered the only possible or economically optimal solution. However, the equipment used for decapsulation was cost-effective compared to professional ones. While the total cost of our setup was approximately €7,000, **specialized commercial decapsulation systems typically start at €50,000 and above**.

Since the bulk material removal by laser does not critically depend on extreme power or sub-micron precision, comparable results can be achieved with significantly lower expenses—for example, by utilizing standard laser engravers priced from €500 - €700. The critical factor remains identifying the optimal combination of scan speed and laser power to prevent thermal damage to the silicon die. This parameter optimization process is more

straightforward when accessing the top side of the package. For the backside, mechanical milling using an engraving router appears to be a more suitable alternative, as low-cost laser systems with slower scan speeds tend to melt the copper thermal pad rather than ablate it effectively.

The crucial phase of precision work in close proximity to the die surface is primarily a matter of chemical processes. During chemical application, optimizing the exposure time and precisely controlling the reaction temperature are paramount. Even in this step, a laboratory hot plate is not a strict requirement; any alternative heat source capable of stably maintaining a surface temperature between 60–90 °C can be utilized. A similar situation applies to the digital inspection microscope or the chemical dosing tool. With these devices, appropriate results can also be achieved using commonly available alternatives at a fraction of the procurement cost. However, it must be noted that implementing these low-cost solutions will undoubtedly be more labor-intensive and may lead to a lower repeatability rate of the overall process. The absolute financial baseline required to establish a functional setup using these alternatives can be roughly estimated at approximately €2,000.

5 Conclusion

The following key factors were identified as critical for the success of both decapsulation procedures:

- optimization of laser ablation parameters (power, scan speed, repetitions, and resolution) to enable efficient material removal without risking thermal damage to the device;
- characterization of the chemical process, including the optimal acid mixture composition, concentration, operating temperature, and exposure time for selective etching of the molding compound and copper thermal pad;
- development of a protective methodology for PCB solder pads and package leads against corrosion during etching and the definition of a post-etch rinsing procedure to maximize the lifespan of the exposed components;
- establishment of an effective method for removing conductive silver die-attach epoxy from the backside of the die without inducing mechanical defects or thermal stress to the silicon surface.

Resolving these critical areas has enabled successful exposure of the silicon die, ensuring high reliability and repeatability throughout the procedure.

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